Independence

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Definition 0.1 (Statistical independence)**.** Random variables $X_1, \ldots, X_n$ are **statistically independent** if, for all sets of real numbers $A_1, \ldots, A_n$, the probability that every X_i falls in its set is the product of the individual probabilities:

$$\Pr(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \Pr(X_i \in A_i)$$

Remark. We write $X \perp\!\!\!\perp Y$ for "X and Y are independent". The symbol $\perp\!\!\!\perp$ is essentially $\prod$ upside-down, which can remind you of the definition.

Theorem 0.1 (Independence of discrete random variables: the joint PMF factors)**.** *Discrete random variables $X_1, \ldots, X_n$ are statistically independent if and only if their joint PMF factors into the product of their PMFs: for all real numbers $x_1, \ldots, x_n$,*

$$\mathrm{P}(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} \mathrm{P}(X_i = x_i)$$

Proof

Proof. **Only if.** Take $A_i = \{x_i\}$ for each i in Definition 0.1.
If. Let $A_1, \ldots, A_n$ be sets of real numbers, and let C be the set of tuples $(x_1, \ldots, x_n)$ with each $x_i \in A_i \cap \mathcal{R}(X_i)$. Each range $\mathcal{R}(X_i)$ is countable, so C is countable. Each X_i takes its values in $\mathcal{R}(X_i)$, so the event $\{X_1 \in A_1, \ldots, X_n \in A_n\}$ is the disjoint union of the events $\{X_1 = x_1, \ldots, X_n = x_n\}$ over $(x_1, \ldots, x_n) \in C$:

$$
\begin{aligned}
\Pr(X_1 \in A_1, \dots, X_n \in A_n) &= \sum_{(x_1,\dots,x_n)\in C} \mathrm{P}(X_1 = x_1, \dots, X_n = x_n) && \text{(countable additivity over the disjoint even} \\
&= \sum_{(x_1,\dots,x_n)\in C} \prod_{i=1}^{n} \mathrm{P}(X_i = x_i) && \text{(the joint PMF factors)} \\
&= \prod_{i=1}^{n} \sum_{x_i \in A_i \cap \mathcal{R}(X_i)} \mathrm{P}(X_i = x_i) && \text{(a sum over the product set } C \text{ of non-nega} \\
&= \prod_{i=1}^{n} \Pr(X_i \in A_i) && \text{(countable additivity over the disjoint even}
\end{aligned}
$$

□

Example 0.1 (Two fair coin flips)**.** Flip two fair coins, and let X_1 and X_2 indicate heads on the first and second flip. Each of the four outcomes has probability $1/4$, so for example:

$$
\begin{aligned}
\mathrm{P}(X_1 = 1, X_2 = 1) &= \tfrac{1}{4} \\
&= \tfrac{1}{2} \cdot \tfrac{1}{2} \\
&= \mathrm{P}(X_1 = 1)\ \mathrm{P}(X_2 = 1)
\end{aligned}
$$

and the same factorization holds for the other three pairs of values, so $X_1 \perp\!\!\!\perp X_2$ by Theorem 0.1. In contrast, X_1 and the total number of heads, $X_1 + X_2$, are not independent: $\mathrm{P}(X_1 = 0, X_1 + X_2 = 2) = 0$, but

$$
\begin{aligned}
\mathrm{P}(X_1 = 0)\ \mathrm{P}(X_1 + X_2 = 2) &= \tfrac{1}{2} \cdot \tfrac{1}{4} \\
&= \tfrac{1}{8}.
\end{aligned}
$$

Theorem 0.2 (Independence with densities: the joint density factors)**.** ***Continuous case.*** *Continuous random variables X and Y with densities $\mathrm{p}(X = x)$ and $\mathrm{p}(Y = y)$ are statistically independent if and only if the product $\mathrm{p}(X = x)\ \mathrm{p}(Y = y)$ is a joint density of X and Y.*
Mixed case. *A discrete random variable X and a continuous random variable Y with density $\mathrm{p}(Y = y)$ are statistically independent if and only if the product $\mathrm{P}(X = x)\ \mathrm{p}(Y = y)$ is a joint density-mass function of X and Y.*

Remark. The proof is beyond these notes' scope: it needs measure theory that the notes do not develop, to extend a density's integrals from intervals to general sets and to exchange the order of integration (Casella and Berger 2002; Billingsley 1995). The continuous case extends to n random variables with a joint density of all n. The statement says "a joint density" rather than "the joint density" because a density is not unique: changing it on a

set of zero area (zero length, in the mixed case) changes none of its integrals.

Example 0.2 (The PMF form fails for continuous random variables)**.** The factorization in Theorem 0.1 does not work as a definition of independence for continuous random variables: there, both sides are 0 at every point, so it would call every pair of continuous random variables independent. For instance, let $X \sim \text{Uniform}(0,1)$ (uniform distribution), whose density is 1 on $[0,1]$, and let $Y = X$. For all real numbers x and y, the event $\{X = x,\, Y = y\}$ is $\{X = x\}$ if $y = x$ and empty otherwise, so it has probability 0, because $\Pr(X = x) = 0$ for the continuous X. Likewise

$$\begin{aligned}\Pr(X = x)\ \Pr(Y = y) &= 0 \cdot 0 \\ &= 0,\end{aligned}$$

so the PMF form holds. But X and Y are not independent:

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}],\, Y \in [0, \tfrac{1}{2}]) &= \Pr(X \in [0, \tfrac{1}{2}]) && (Y = X) \\ &= \int_0^{1/2} 1\, dx && (\text{the density of } X \text{ is 1 on } [0,1]) \\ &= \tfrac{1}{2} && (\text{integrate})\end{aligned}$$

while

$$\begin{aligned}\Pr(X \in [0, \tfrac{1}{2}])\ \Pr(Y \in [0, \tfrac{1}{2}]) &= \tfrac{1}{2} \cdot \tfrac{1}{2} \\ &= \tfrac{1}{4}.\end{aligned}$$

Definition 0.2 (Conditional independence)**.** Random variables $Y_1, \ldots, Y_n$ are **conditionally independent** given a discrete random variable (or vector) $\tilde{X}$ if, for every value $\tilde{x}$ with $\Pr(\tilde{X} = \tilde{x}) > 0$ and all sets of real numbers $A_1, \ldots, A_n$, the conditional probability that every Y_i falls in its set is the product of the individual conditional probabilities:

$$\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) = \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x})$$

When $\tilde{X}$ is continuous, every event $\{\tilde{X} = \tilde{x}\}$ has probability 0, so the condition is stated with densities instead: for every $\tilde{x}$ with $\mathrm{p}(\tilde{X} = \tilde{x}) > 0$ and all $y_1, \ldots, y_n$,

$$\frac{\mathrm{p}(\tilde{X} = \tilde{x}, Y_1 = y_1, \ldots, Y_n = y_n)}{\mathrm{p}(\tilde{X} = \tilde{x})} = \prod_{i=1}^{n} \frac{\mathrm{p}(\tilde{X} = \tilde{x}, Y_i = y_i)}{\mathrm{p}(\tilde{X} = \tilde{x})}$$

where each $\mathrm{p}(\cdot)$ is a joint density (with probability masses in place of densities for any discrete Y_i, as in a joint density-mass function).

Remark. For continuous $\tilde{X}$, each ratio in the density form is the density of the Y's under their conditional distribution given $\tilde{X} = \tilde{x}$, so both forms say the same thing: given $\tilde{X} = \tilde{x}$, the Y_i's joint distribution factors.

Example 0.3 (Two tests of the same patient)**.** Let X indicate whether a patient has a disease, and let Y_1 and Y_2 indicate positive results on two tests whose errors are unrelated, so that Y_1 and Y_2 are conditionally independent given X. Suppose $\Pr(X = 1) = 0.1$, each test is positive with probability 0.9 if $X = 1$ and with probability 0.1 if $X = 0$. Then, by the law of total probability:

$$\begin{aligned}
\Pr(Y_1 = 1, Y_2 = 1) &= (0.9)(0.9)(0.1) + (0.1)(0.1)(0.9) && \text{(condition on } X\text{; factor given } X\text{)} \\
&= 0.081 + 0.009 && \text{(multiply)} \\
&= 0.09 && \text{(add)}
\end{aligned}$$

but

$$\begin{aligned}
\Pr(Y_1 = 1) &= (0.9)(0.1) + (0.1)(0.9) \\
&= 0.18,
\end{aligned}$$

so $\Pr(Y_1 = 1)\ \Pr(Y_2 = 1) = 0.0324 \neq 0.09$: the tests are conditionally independent given X, but not independent.

Example 0.4 (Two coins, given their total)**.** Let Y_1 and Y_2 indicate heads on two fair coin flips, which are independent (Example 0.1), and let $X = Y_1 + Y_2$ be the number of heads. Given $X = 1$, which has probability $1/2$, exactly one coin is heads, so:

$$\begin{aligned}
\Pr(Y_1 = 1, Y_2 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, Y_2 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{0}{1/2} && \text{(two heads make } X = 2\text{, not 1)} \\
&= 0 && \text{(divide)}
\end{aligned}$$

but

$$
\begin{aligned}
\Pr(Y_1 = 1 \mid X = 1) &= \frac{\Pr(Y_1 = 1, X = 1)}{\Pr(X = 1)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(Y_1 = 1, Y_2 = 0)}{\Pr(X = 1)} && (Y_1 = 1 \text{ and } X = 1 \text{ means } Y_2 = 0) \\
&= \frac{1/4}{1/2} && \text{(substitute)} \\
&= \tfrac{1}{2} && \text{(divide)}
\end{aligned}
$$

and likewise $\Pr(Y_2 = 1 \mid X = 1) = 1/2$, so the product of the conditional probabilities is $1/4 \neq 0$: Y_1 and Y_2 are independent, but not conditionally independent given X.

Proposition 0.1 (Neither independence nor conditional independence implies the other)**.** *There are random variables Y_1, Y_2, and X such that Y_1 and Y_2 are conditionally independent given X but not independent, and there are random variables Y_1, Y_2, and X such that Y_1 and Y_2 are independent but not conditionally independent given X.*

Proof

Proof. Example 0.3 gives the first case, and Example 0.4 gives the second. □

Example 0.5 (Wet roads, umbrellas, and rain)**.** Consider Brian Hutchinson's intuitive illustration of conditional independence (Hutchinson 2024):
Let w denote wet roads, u denote people carrying umbrellas, and r denote rain.

- **Marginally, wet roads and umbrellas are not independent ($w \not\perp u$).** If you look out a window and see people holding open umbrellas, that increases the probability that the pavement outside is wet.
- **Conditionally on rain, they are independent ($w \perp u \mid r$).** Once you know with certainty whether it is currently raining, learning whether pedestrians are carrying umbrellas tells you nothing more about the state of the roads:

$$
\Pr(w, u \mid r) = \Pr(w \mid r)\ \Pr(u \mid r), \qquad \Pr(w \mid u, r) = \Pr(w \mid r)
$$

Proposition 0.2 (Conditional independence when each Y_i depends only on its own X_i)**.** *Let $\tilde{X} = (X_1, \ldots, X_n)$ be a discrete random vector, and let $Y_1, \ldots, Y_n$ be conditionally independent given $\tilde{X}$. Suppose also that each Y_i depends on $\tilde{X}$ only through X_i: for every $\tilde{x} = (x_1, \ldots, x_n)$ with $\Pr(\tilde{X} = \tilde{x}) > 0$ and every set of real numbers A_i,*

$$\Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x}) = \Pr(Y_i \in A_i \mid X_i = x_i)$$

Then, for every such $\tilde{x}$ and all sets of real numbers $A_1, \ldots, A_n$:

$$\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) = \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid X_i = x_i)$$

Proof

Proof. Each conditional probability given $X_i = x_i$ is defined, because $\Pr(X_i = x_i) \geq \Pr(\tilde{X} = \tilde{x}) > 0$. So:

$$\begin{aligned}
\Pr(Y_1 \in A_1, \ldots, Y_n \in A_n \mid \tilde{X} = \tilde{x}) &= \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid \tilde{X} = \tilde{x}) && \text{(conditional independence given } \tilde{X}) \\
&= \prod_{i=1}^{n} \Pr(Y_i \in A_i \mid X_i = x_i) && \text{(each } Y_i \text{ depends on } \tilde{X} \text{ only through } X_i)
\end{aligned}$$

□

Remark. In regression, $\tilde{X}$ is usually the full set of covariates $(X_1, \ldots, X_n)$, and a model usually also assumes, as Proposition 0.2 does, that each Y_i depends on the covariates only through its own X_i. That second assumption is a separate one, not part of conditional independence.

Definition 0.3 (Identically distributed)**.** Random variables $X_1, \ldots, X_n$ are **identically distributed** if they all have the same CDF:

$$\forall i \in \{1, \ldots, n\}, \forall x \in \mathbb{R} : \Pr(X_i \leq x) = \Pr(X_1 \leq x)$$

Theorem 0.3 (Identically distributed discrete random variables have the same PMF)**.** *Discrete random variables $X_1, \ldots, X_n$ are identically distributed if and only if they all have the same PMF: for every $i \in \{1, \ldots, n\}$ and every real number x,*

$$\mathrm{P}(X_i = x) = \mathrm{P}(X_1 = x)$$

Proof

Proof. Write $F_i(t) \stackrel{\text{def}}{=} \Pr(X_i \le t)$ for the CDF of X_i.

Only if: the CDF determines the PMF through its jumps. Fix i and x. Every number below x lies in exactly one of the intervals $(-\infty, x-1]$, $(x-1, x-\frac{1}{2}]$, $(x-\frac{1}{2}, x-\frac{1}{3}]$, ..., so the event $\{X_i < x\}$ is the disjoint union of the events $B_1 \stackrel{\text{def}}{=} \{X_i \le x-1\}$ and $B_k \stackrel{\text{def}}{=} \{x - \frac{1}{k-1} < X_i \le x - \frac{1}{k}\}$ for $k \ge 2$, and for each K, the events $B_1, \ldots, B_K$ have union $\{X_i \le x - \frac{1}{K}\}$:

$$\begin{aligned}
\Pr(X_i < x) &= \sum_{k=1}^{\infty} \Pr(B_k) && \text{(countable additivity)} \\
&= \lim_{K\to\infty} \sum_{k=1}^{K} \Pr(B_k) && \text{(definition of an infinite series)} \\
&= \lim_{K\to\infty} \Pr(X_i \le x - \tfrac{1}{K}) && \text{(finite additivity)} \\
&= \lim_{K\to\infty} F_i(x - \tfrac{1}{K}) && \text{(definition of the CDF)}
\end{aligned}$$

The event $\{X_i \le x\}$ is the disjoint union of $\{X_i < x\}$ and $\{X_i = x\}$, so:

$$\begin{aligned}
\mathrm{P}(X_i = x) &= \Pr(X_i \le x) - \Pr(X_i < x) && \text{(finite additivity)} \\
&= F_i(x) - \Pr(X_i < x) && \text{(definition of the CDF)} \\
&= F_i(x) - \lim_{K\to\infty} F_i(x - \tfrac{1}{K}) && \text{(the display above)} \\
&= F_1(x) - \lim_{K\to\infty} F_1(x - \tfrac{1}{K}) && (F_i = F_1) \\
&= \mathrm{P}(X_1 = x) && \text{(the same two steps, for } X_1)
\end{aligned}$$

If: the PMF determines the CDF through sums. Let S be the set of values u with $\mathrm{P}(X_1 = u) > 0$. Since the PMFs are equal, S is also the set of u with $\mathrm{P}(X_i = u) > 0$, so $S \subseteq \mathcal{R}(X_i)$ and $S \subseteq \mathcal{R}(X_1)$. For every real t, the event $\{X_i \le t\}$ is the disjoint union of the events $\{X_i = u\}$ over the countably many $u \in \mathcal{R}(X_i)$ with $u \le t$, so:

$$\begin{aligned}
F_i(t) &= \sum_{u \in \mathcal{R}(X_i),\, u \le t} \mathrm{P}(X_i = u) && \text{(countable additivity)} \\
&= \sum_{u \in S,\, u \le t} \mathrm{P}(X_i = u) && \text{(drop the terms that are 0)} \\
&= \sum_{u \in S,\, u \le t} \mathrm{P}(X_1 = u) && \text{(the PMFs are equal)} \\
&= \sum_{u \in \mathcal{R}(X_1),\, u \le t} \mathrm{P}(X_1 = u) && \text{(restore the terms that are 0)} \\
&= F_1(t) && \text{(countable additivity)}
\end{aligned}$$

□

Remark. As with independence (Example 0.2), the PMF form would not work as a definition in general: every continuous random variable has $\Pr(X_i = x) = 0$ at every x.

Example 0.6 (Identically distributed but not independent)**.** In Example 0.1, X_1 and $1 - X_1$ (the indicator of tails on the first flip) both take the values 0 and 1 with probability $1/2$ each, so they are identically distributed (Theorem 0.3). They are not independent: knowing one determines the other.

Definition 0.4 (Conditionally identically distributed)**.** Random variables $Y_1, \ldots, Y_n$ are **conditionally identically distributed** given random variables $X_1, \ldots, X_n$ if the conditional CDF of each Y_i given $X_i = x$ is one shared function $G(y \mid x)$ of y and x: for every $i \in \{1, \ldots, n\}$, every y, and every x with $\Pr(X_i = x) > 0$,

$$\Pr(Y_i \leq y \mid X_i = x) = G(y \mid x)$$

When X_i is continuous, the condition applies at every x with $\mathrm{p}(X_i = x) > 0$, with $\Pr(Y_i \leq y \mid X_i = x)$ computed from densities: the integral of $\mathrm{p}(X_i = x, Y_i = u)/\mathrm{p}(X_i = x)$ over $u \leq y$ (a sum over values $u \leq y$ when Y_i is discrete).

Example 0.7 (A shared regression model)**.** Suppose each Y_i is binary, with $\Pr(Y_i = 1 \mid X_i = x) = \pi(x)$ for one function π shared by every i (for instance, $\pi(x) = x/(1 + x)$ for a dose $x \geq 0$). Then $Y_1, \ldots, Y_n$ are conditionally identically distributed given $X_1, \ldots, X_n$, with $G(y \mid x) = 1 - \pi(x)$ for $0 \leq y < 1$ (and 0 for $y < 0$, 1 for $y \geq 1$). Their marginal distributions can still differ: if participant 1 always receives dose 0 and participant 2 always receives dose 1, then $\Pr(Y_1 = 1) = 0$ but $\Pr(Y_2 = 1) = 1/2$.

Definition 0.5 (Independent and identically distributed)**.** Random variables $X_1, \ldots, X_n$ are **independent and identically distributed** (shorthand: "X_i iid") if they are:

- statistically independent, and
- identically distributed.

Remark. The IID assumption is one of the most common assumptions in introductory statistics: it says a sample $X_1, \ldots, X_n$ can be treated as n independent draws from a single shared distribution.

Example 0.8 (Repeated die rolls)**.** The results of n rolls of the same fair die are IID: the rolls are independent, and each is uniform on $\{1, \ldots, 6\}$.

Definition 0.6 (Conditionally independent and identically distributed)**.** Random variables $Y_1, \dots, Y_n$ are **conditionally independent and identically distributed** given $X_1, \dots, X_n$ (shorthand: "$Y_i \mid X_i$ ciid" or just "$Y_i \mid X_i$ iid") if:

- $Y_1, \dots, Y_n$ are conditionally independent given $(X_1, \dots, X_n)$,
- each Y_i depends on $(X_1, \dots, X_n)$ only through X_i, and
- $Y_1, \dots, Y_n$ are conditionally identically distributed given $X_1, \dots, X_n$.

Example 0.9 (The usual regression assumption)**.** In Example 0.7, if the Y_i are also conditionally independent given all the doses, and each Y_i depends on the doses only through its own X_i, then $Y_i \mid X_i$ ciid, and the joint conditional PMF is $\prod_{i=1}^{n} \pi(x_i)^{y_i}(1-\pi(x_i))^{1-y_i}$: one shared function evaluated at each (x_i, y_i).

Video lecture

Hutchinson's Probability Refresher (27 min) covers independence (Hutchinson, n.d.). The login for the video site is posted on Canvas.

Another explanation

5MinuteAI explains independence in five minutes, following the University of Washington's CSE 312 (5MinuteAI, n.d.).

References

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